MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL TTL, LOW-POWER, MULTIPLE NOR GATES, MONOLITHIC SILICON

This limited coordination Military specification has been prepared by the Air Force Space and Missile Systems Organization based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-M-38510/27(USAF). It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1. Scope. This specification covers the requirements for monolithic silicon, TTL, low-power, positive NOR logic gating microcircuits for use in Air Force missile and space systems.

1.2 Part number. The complete part number shall be as shown in the following example:

M38510	/027	01	S	T	D
Military designator	Detail specification	Device type (1.2.1)	Device class (1.2.2)	Case outline (1.2.3)	Lead finish (3.3)

1.2.1 Device type. The device type shall be as shown in the following:

Device type	Circuit	Similar commercial type
01	Quadruple 2-input positive NOR gate	54L02

1.2.2 Device class. The device class shall be Class S as defined in MIL-M-0038510.

1.2.3 Case outline. The case outline shall be as specified in figure 10 of MIL-M-0038510 and shall be identified as outline T.

1.3 Absolute maximum ratings.

Supply voltage range	0 Vdc to 8.0 Vdc
Input voltage range	0 Vdc to 5.5 Vdc
Storage temperature range	-65°C to 150°C
Lead temperature (soldering, 10 seconds)	300°C

1.4 Recommended operating conditions.

Ambient operating temperature range	-55°C to 125°C
Supply voltage	4.5 Vdc minimum to 5.5 Vdc maximum
Minimum high level input voltage	2.0 Vdc
Maximum low level input voltage	0.7 Vdc

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2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-0038510 - Microcircuits, General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-0038510 and as specified herein.

3.2 Design, construction, weight and physical dimensions. The design, construction, weight and physical dimensions shall be as specified in MIL-M-0038510 and herein.

3.2.1 Logic diagram and terminal connections. The logic diagram and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.3 Schematic circuit. The schematic circuit shall be as specified on figure 3.

3.2.4 Case outline. The case outline shall be as specified in 1.2.3.

3.2.5 Weight. The maximum weight shall be 1 gram.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-0035810.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in tables I and II.

3.4.1 Delta drift limits. The delta drift limits shall be as specified in table I herein.

3.5 Screening. Screening shall be in accordance with table I of MIL-M-0038510 and as specified herein.

3.5.1 Interim electrical parameters. The interim electrical parameters shall consist of all the electrical characteristics tests as specified in table I. Each test shall be performed go, no-go at -55°C and 125°C. Each test shall also be read and recorded at 25°C. For all tests, the devices shall meet the limits specified in table I.

3.5.2 Final electrical tests. The final electrical tests shall consist of all the electrical characteristics tests specified in table I. Each test shall be read and recorded at -55°C, 25°C and 125°C. For all tests, the devices shall meet the limits specified in table I.

3.5.2.1 Delta computation. Delta computations shall be performed for those parameters in table I which are marked with an asterisk. All devices shall meet the applicable delta limits specified in table I. (See section 6 in MIL-M-0038510.)

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TABLE I. Electrical performance characteristics.

1/ 2/ TEST NUMBER	TEST NAME	PIN NUMBERS														3/ LIMITS		1/ DELTA Δ	UNITS
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	MIN	MAX		
1*	V_{OH}	0.7 V	0.7 V	-100 µA	4.5 V							GND				2.4		±0.12	V
2*	↑				↑	-100 µA	0.7 V	0.7 V				↑				↑		↑	↑
3*	↓								-100 µA	0.7 V	0.7 V					↓		↓	
4*	V_{OH}												0.7 V	0.7 V	-100 µA	2.4		±0.12	
5*	V_{OL}	GND	2 V	2.0 mA													0.3	±0.03	
6*	↑					2.0 mA	2 V	GND									↑	↑	
7*		2 V	GND	2.0 mA															
8*						2.0 mA	GND	2 V											
9*									2.0 mA	GND	2 V								
10*													GND	2 V	2.0 mA				
11*	↓				↓				2.0 mA	2 V	GND						↓	↓	↓
12*	V_{OL}				4.5 V								2 V	GND	2.0 mA		0.3	±0.03	V
13	I_{IL}	4.5 V	0.3 V		5.5 V												-180		µA
14	↑				↑		0.3 V	4.5 V									↑		↑
15		0.3 V	4.5 V																
16							4.5 V	0.3 V											
17										4.5 V	0.3 V								
18													0.3 V	4.5 V					
19	↓									0.3 V	4.5 V						↓		
20	I_{IL}												4.5 V	0.3 V			-180		
21*	I_{IH1}	GND	2.4 V									↓					10	10X or 1 µA	↓
22*	I_{IH1}				5.5 V		GND	2.4 V				GND					10	10X or 1 µA	µA

See notes at the end of this table.

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TABLE I. Electrical performance characteristics. - Continued

1/2/ TEST NUMBER	TEST NAME	PIN NUMBERS														3/ LIMITS		1/ DELTA Δ	UNITS
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	MIN	MAX		
23*	I_{IH1}	2.4 V	GND		5.5 V							GND					10	10X or 1 µA	µA
24*	↑				↑		2.4 V	GND				↑					↑	↑	↑
25*										GND	2.4 V								
26*													GND	2.4 V					
27*	↓									2.4 V	GND						↓	↓	
28*	I_{IH1}												2.4 V	GND			10	10X or 1 µA	
29	I_{IH2}	GND	5.5 V														100		
30	↑						GND	5.5 V									↑		
31		5.5 V	GND																
32							5.5 V	GND											
33										GND	5.5 V								
34													GND	5.5 V					
35	↓									5.5 V	GND						↓		↓
36	I_{IH2}												5.5 V	GND			100		µA
37	I_{OS}	GND	GND	0.0 V												-3.0	-15.0		mA
38	↑					0.0 V	GND	GND								↑	↑		↑
39	↓								0.0 V	GND	GND					↓	↓		
40	I_{OS}												GND	GND	0.0 V	-3.0	-15.0		
41	I_{PD}	5.0 V	GND		↓		5.0 V	GND		5.0 V	GND	↓	5.0 V	GND			2.6		↓
42	I_{PD}	GND	GND		5.5 V		GND	GND		GND	GND	GND	GND	GND			1.6		mA

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurements"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

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Device type 01

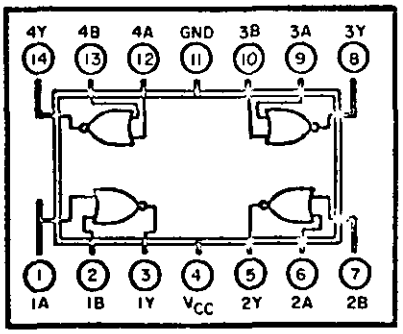


FIGURE 1. Logic diagram and terminal connections.

Device type 01

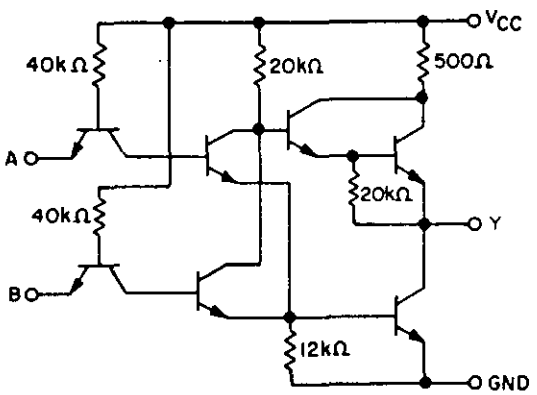
Truth table each gate		
Input		Output
A	B	Y
H	X	L
X	H	L
L	L	H

X = Irrelevant

Positive logic: $Y = \overline{A + B}$

FIGURE 2. Truth tables and logic equations.

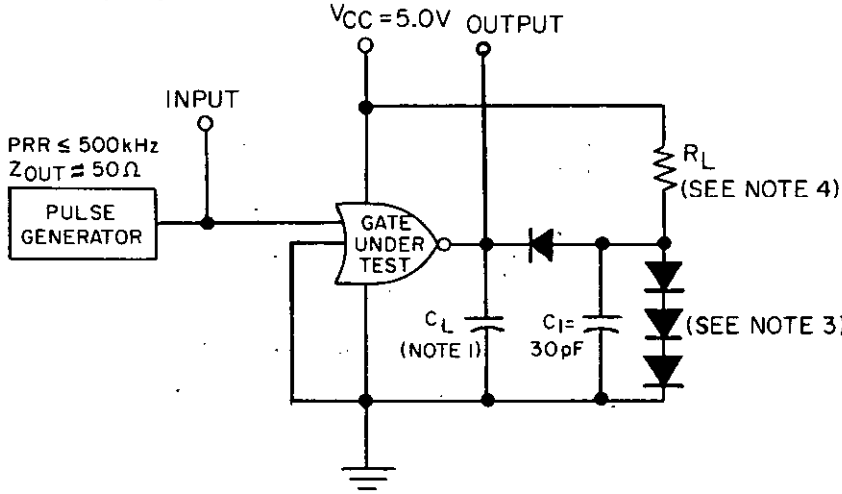
Device type 01

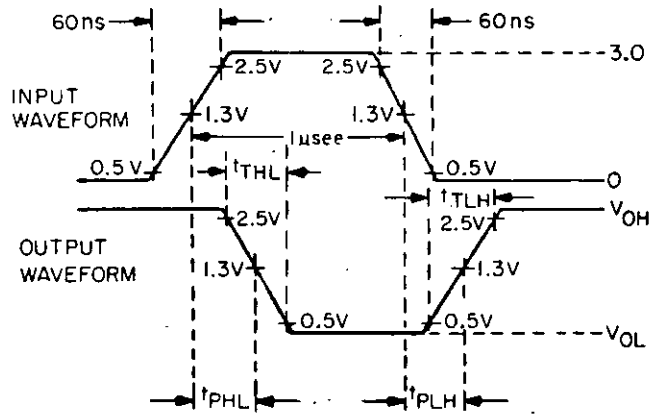


NOTE: Component values shown are nominal.

FIGURE 3. Schematic circuit.

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Device type 01

NOTES:
1. C_L = 50 pF minimum including jig and probe capacitance.
2. Propagation Delay = $\frac{t_{PHL} + t_{PLH}}{2}$.
3. All diodes are 1N916 or equivalent.
4. R_L = 4 kΩ ±5%.

FIGURE 4. Switching time test circuit and waveform.

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3.5.3 Switching characteristics tests. The switching characteristics tests specified in table II and figure 4 shall be performed at 25°C. All devices shall meet the limits specified in table II.

3.6 Extended life tests. Extended life tests shall be performed in accordance with MIL-M-0038510.

3.7 Marking. Marking shall be in accordance with MIL-M-0038510 and 1.2 herein.

4. PRODUCT ASSURANCE PROVISIONS.

4.1 Product assurance provisions. Product assurance shall be in accordance with MIL-M-0038510.

TABLE II. Switching characteristics. 1/

Characteristic	Symbol	Conditions	Test figure	Limits		Units
				Min	Max	
Propagation delay time to low level output	t_{PHL}	C_L = 50 pF, R_L = 4 kΩ, T_A = 25°C, V_{CC} = 5 V	Figure 4	-	60	ns
Propagation delay time to high level output	t_{PLH}			-	60	ns

1/ Read and record measurements.

5. PREPARATION FOR DELIVERY.

5.1 Preparation for delivery. Preparation for delivery shall be as specified in MIL-M-0038510.

6. NOTES.

6.1 Notes. Notes are as specified in MIL-M-0038510.

Custodian:
Air Force - 19

Review activities:
Air Force - 17, 19
DSA - ES

Preparing activity:
Air Force - 19

Agent:
DSA - ES

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